

BMEG3105 · Term 1 2026–27

Lecture 6

Visualization and Clustering Concepts

Expanded concept and example edition

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Continuation of Lecture 5: finish exploratory visualization, introduce clustering, and quantify similarity and distance. Sections 6.1–6.9 match the companion working notes. Examples marked Supplementary extend the classroom concepts.

6.1 Reading distributions and relationships

6.1.1 Histogram bins change what we can see

The lecture resumes with a practical question: once a data matrix has been cleaned, how can we understand its distribution? A histogram groups a numerical attribute into intervals and counts observations in each interval. It shows where observations accumulate, where gaps occur, and where an isolated value may need checking. The classroom example uses Iris petal width. [1, pp. 11–13]

Too few bins can conceal structure; too many narrow bins can produce sparse, unstable-looking counts. Distinguish bin width from the number of bins: over a fixed range, narrower bins mean more bins. The relevant choice is a resolution that reveals useful structure without treating every small fluctuation as meaningful.

Supplementary example S1 One dataset, three views. Use 1, 1, 2, 2, 3, 3, 4, 4, 8, 9. A single bin from 0 to 10 contains all ten values. Five width-2 bins have counts 2, 4, 2, 0, 2. Ten width-1 bins expose finer detail, including several empty intervals.

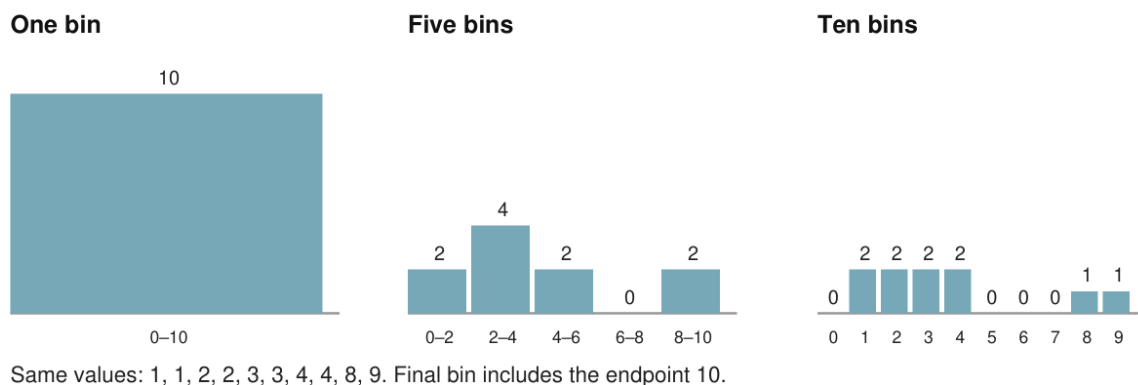


Figure 6.1 Constructed data under three bin choices. Intervals include the left endpoint and exclude the right, except for the final bin, which includes both endpoints.

The single bar hides the gap between the lower and higher values. Five bins make the empty interval [6,8) visible. Ten bins give a more fragmented view without adding observations. Choosing a binning changes the display, not the measurements.

Classroom connection. A height near 11 m, far from the usual 1–2 m values, is easy to notice in a histogram. The plot flags a possible data-quality problem; the units and original entry still need checking before a value is changed.

6.1.2 A joint distribution preserves the pairing

The two-dimensional histogram adds a second measured attribute. In the classroom Iris example, the horizontal axes represent petal length and petal width, and bar height represents the number of flowers in each pair of intervals. A tall bar therefore means many observations have that particular combination of length and width. [1, p. 14]

Counts concentrated along a rising diagonal suggest that larger values of one attribute tend to occur with larger values of the other. The lecture uses this visual pattern to motivate correlation. A plot suggests a relationship; it does not by itself establish a numerical correlation coefficient or a causal explanation.

Supplementary example S2 Count pairs, not separate columns. Consider six hypothetical flowers with (length, width) values (1,1), (1.5,1.5), (2.5,2.5), (3,3), (4.5,4.5), and (5,5). Use the intervals [0,2), [2,4), and [4,6] for both attributes. Each diagonal cell contains two flowers, and every other cell contains none.

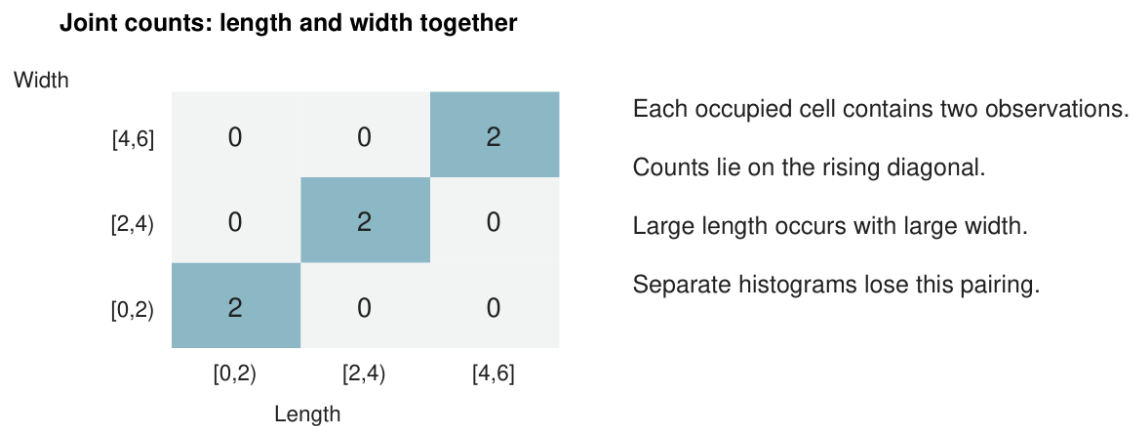


Figure 6.2 A two-dimensional count grid for the constructed pairs. A shaded cell represents the same count information that a bar would encode in a three-dimensional histogram.

For instance, (2.5,2.5) and (3,3) both enter the middle cell because both coordinates are in [2,4). Summing each column gives a length histogram with counts 2,2,2. Summing each row gives the same width histogram. These marginal summaries omit which width belongs to which length.

Why the pairing matters. Reverse the width values while keeping the lengths in order: the occupied cells move to the falling diagonal. Both separate histograms remain unchanged, but the relationship reverses. Two datasets can therefore have identical one-variable distributions and very different joint patterns.

Choose attributes → preserve paired observations → count cells → interpret the pattern

This connects exploration to later similarity calculations: both depend on how attributes are paired and represented. A data matrix is more than a collection of unrelated columns.

6.2 Comparing distributions with box plots

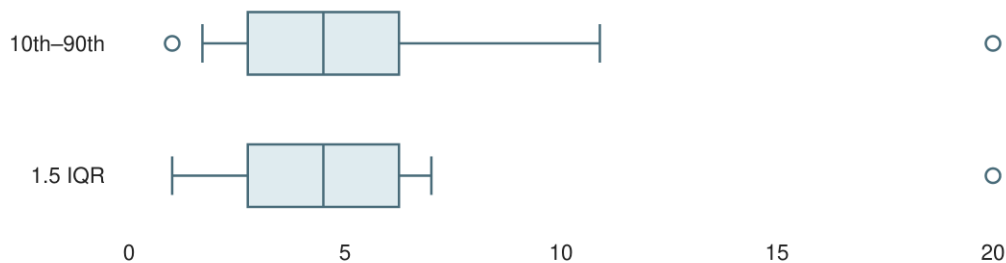
A box plot summarizes location and spread using selected percentiles. The box runs from the 25th percentile (Q1) to the 75th percentile (Q3); the line inside marks the median. Comparing boxes shows differences in their central values and in the spread of their middle observations. [1, p. 15]

Read the convention before interpreting the whiskers. The classroom schematic explicitly labels the lower and upper whiskers as the 10th and 90th percentiles. Other box plots use different rules, including whiskers extending to observed values within $1.5 \times \text{IQR}$ of the quartiles. These conventions must not be silently interchanged. [3]

Supplementary example S3 Work through both rules. Take sorted values 1, 2, 3, 4, 5, 6, 7, 20. For this example, define the p-quantile by linear interpolation at zero-based position $(n-1)p$. Then Q1 is at position 1.75, giving 2.75; the median is 4.5; and Q3 is at position 5.25, giving 6.25.

Quantity	Calculation	Value
IQR	$6.25 - 2.75$	3.5
10th percentile	$1 + 0.7(2 - 1)$	1.7
90th percentile	$7 + 0.3(20 - 7)$	10.9
Lower 1.5 IQR fence	$2.75 - 1.5 \times 3.5$	-2.5
Upper 1.5 IQR fence	$6.25 + 1.5 \times 3.5$	11.5

The same data under two stated whisker rules



Data: 1, 2, 3, 4, 5, 6, 7, 20. Quartiles use linear interpolation in this example.

Figure 6.3 The box and median are unchanged, but the whisker rule changes which observations lie beyond the whiskers. All values are constructed for this comparison.

With percentile whiskers, 1 and 20 lie outside 1.7–10.9. With the 1.5 IQR rule, the whiskers end at observed values 1 and 7, and 20 is flagged. A point beyond a whisker is a signal to investigate, not proof of a measurement error or a clinical abnormality.

The lecture compares relatively concentrated and widely spread Iris features. Check that comparisons use meaningful units and remember that the box summarizes the middle portion of a distribution, not every detail of its shape.

6.3 Why group objects into clusters

Clustering finds groups whose members are similar to one another and different from members of other groups. The lecture expresses this as small intra-cluster differences and large inter-cluster differences. It is a useful organizing objective, although different algorithms formalize it in different ways. [1, pp. 18–21]



Figure 6.4 A schematic of three separated groups. Colors identify the illustrated groups; the geometry supplies the similarity pattern.

Classroom example	What is grouped	Why it helps
Books or supermarket items	Items with related subjects or uses	Organization and retrieval
Customers	People with related interests or needs	Match products to relevant groups
Genes	Expression profiles across conditions	Explore co-expression patterns
Samples or cells	Profiles measured for each sample or cell	Explore candidate subgroups

The biological distinction is important: clustering genes asks which genes behave similarly across samples; clustering samples asks which samples have similar profiles across genes. These are different questions applied to different axes of the same expression matrix.

Supplementary example S4 Similar trends at different levels. Across three conditions, gene A has values (1,2,3), gene B has (2,4,6), and gene C has (3,2,1). A and B increase together even though B has larger measurements. C decreases. A pattern-based comparison can therefore group A with B while a comparison of absolute levels asks a different question.

The lecture motivates investigating disease-related expression and sample subgroups. Clustering generates candidate patterns; it does not by itself perform a differential-expression test, establish a shared biological pathway, or validate a diagnostic category.

Summarization. Replacing a group by representative values can reduce the amount of data to inspect. The lecture also motivates this using patient-group summaries. Aggregation alone does not guarantee privacy: a group containing a single person would reveal that person's value exactly.

6.4 From fruit pictures to a clustering result

The fruit example makes the input and output of clustering explicit. The computer receives numerical data, not an unencoded physical apple or banana. Images can be represented by pixel values, and each image has an identifier such as F1 or F2. A clustering procedure then assigns each item a cluster indicator. [1, pp. 22–29]

Fruit images → numerical features → similarity measure → clustering procedure
 ↓
 cluster indicators → inspect original items

Cluster indicator	Members in the classroom example	What the number means
1	F1, F3, F8	These items are assigned to the same group
2	F2, F4, F6	A second group, not a larger measurement
3	F5, F7, F9	A third group, not a quality ranking

The indicator sequence is 1,2,1,2,3,2,3,1,3. It is not enough simply to obtain these numbers. The lecture returns to the original fruit pictures to check whether objects within each group look alike and whether different groups represent different kinds of fruit.

Supplementary example S5 Labels are names. Rename group 1 as group 3, group 2 as group 1, and group 3 as group 2. The indicators change to 3,1,3,1,2,1,2,3,2, but every pair of items that was together remains together. The partition is unchanged. Cluster numbers have no inherent order or biological identity.

Three ingredients are needed

Ingredient	Clustering	Earlier sequence comparison
Data	Objects expressed as features	Two sequences
Comparison rule	Similarity or distance	A scoring matrix
Procedure	An explicit clustering algorithm	Dynamic programming

A distance measure tells us how to compare two objects; an algorithm tells us what operations to perform to form groups. These roles are connected but not interchangeable. State the representation, comparison rule, and procedure so that the analysis can be reproduced.

Classroom question. Aligning two sequences is not, by itself, clustering. Alignment compares a pair. Its result could become an input to a procedure that groups many sequences. Grouping genes by expression, customers by age, or students by shared interests illustrates the grouping idea.

Terminology: the lecture uses everyday grouping analogies broadly. In formal unsupervised clustering, groups are inferred from the chosen data and method; predefined categories are not automatically a learned clustering solution.

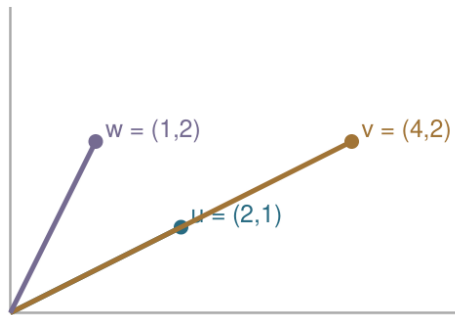
6.5 Similarity and cosine similarity

A similarity score increases as objects become more alike; a dissimilarity or distance decreases. The lecture says similarities often lie between 0 and 1, but this is not a universal range. Cosine similarity and Pearson correlation can also be negative. Always interpret the range and direction of the particular measure. [1, pp. 32–33]

6.5.1 Compare vector directions

For two nonzero vectors u and v , cosine similarity divides their dot product by the product of their Euclidean lengths. It measures the angle between the vectors rather than their overall magnitudes.

$$\cos(u, v) = \frac{u \cdot v}{\|u\|_2 \|v\|_2}$$



u and v have the same direction.

$\cos(u, v) = 1$, despite different lengths.

u and w have different directions.

$\cos(u, w) = 4 / 5 = 0.8$.

Direction and magnitude answer different questions.

Figure 6.5 Supplementary example S6. Multiplying a vector by a positive constant changes its length but preserves its direction and its cosine similarity of 1 with the original.

Work through S6. For $u=(2,1)$ and $v=(4,2)$, the dot product is 10 and the lengths are $\sqrt{5}$ and $\sqrt{20}$. The denominator is also 10, so cosine similarity is 1. For $w=(1,2)$, $u \cdot w=4$ and both lengths are $\sqrt{5}$, giving 0.8. Although u and v have different values, their coordinate proportions agree.

6.5.2 Recalculate the slide example

The slide uses $u=(3,2,0,5,0,0,0,2,0,0)$ and $v=(1,0,0,0,0,0,0,1,0,2)$. Only two positions contribute to the dot product: $3 \times 1 + 2 \times 1 = 5$. The sums of squares are 42 and 6, so the lengths are $\sqrt{42} \approx 6.481$ and $\sqrt{6} \approx 2.449$. Therefore cosine similarity is $5/\sqrt{252} \approx 0.3150$.

Arithmetic clarification: the slide prints 2.245 for $\sqrt{6}$. The correct value is approximately 2.449; the slide's final cosine value of 0.3150 is consistent with the correct square root. [1, p. 33]

For nonnegative count vectors, cosine lies between 0 and 1. With signed vectors, the range is -1 to 1 . A zero vector has no direction, so the formula is undefined. Also distinguish cosine similarity from the software convention “cosine distance,” commonly defined as 1 minus the similarity. [4]

6.6 Correlation measures a linear relationship

Pearson correlation compares how paired values deviate from their respective means. Values near +1 indicate a strong positive linear relationship; values near -1 indicate a strong negative linear relationship. Values near zero indicate little linear association. The lecture illustrates these patterns with scatter plots. [1, pp. 34–35]

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

Classroom example Temperature and ice-cream sales. Calculate each column's mean, subtract that mean, multiply paired deviations, and calculate squared deviations. Sum the three columns and divide the cross-product sum by the square root of the product of the squared-deviation sums. The slide obtains approximately 0.9575, a strong positive linear association.

Recalculation from its twelve displayed observations gives mean temperature 18.675 and mean sales 402.4167. The cross-product sum is 5325.025; the two squared-deviation sums are 176.9825 and 174754.9167. Thus $r = 5325.025 / \sqrt{(176.9825 \times 174754.9167)} \approx 0.9575066$. The slide rounds intermediate values; the conclusion is unchanged.

Supplementary example S7 A small calculation by hand. For $x=(1,2,3)$ and $y=(2,4,6)$, the means are 2 and 4. Center the values before multiplying:

x	y	x - 2	y - 4	Product	(x - 2) ²	(y - 4) ²
1	2	-1	-2	2	1	4
2	4	0	0	0	0	0
3	6	1	2	2	1	4
Sum				4	2	8

The correlation is $4/\sqrt{(2 \times 8)}=1$. Reversing y to (6,4,2) gives -1. Both are strong linear relationships; the sign describes direction. A constant vector has zero centered length, making Pearson correlation undefined.

How it differs from cosine. Correlation is cosine similarity after each vector is centered by its own mean. For $x=(1,2,3)$ and $z=(11,12,13)$, correlation remains 1, whereas raw cosine is $74/\sqrt{6076} \approx 0.949$. Adding a baseline changes the direction from the origin but preserves the centered pattern. [5]

Interpretation: zero correlation does not rule out a nonlinear relationship, and correlation does not establish causation. For example, $x=(-1,0,1)$ and $y=(1,0,1)$ have correlation 0 even though $y=x^2$ exactly.

6.7 Euclidean distance and distance matrices

Euclidean distance is the straight-line distance between two points in a shared feature space. The lecture returns to the need for scaling when different attributes have very different numerical ranges. A distance computed directly from height in meters and weight in kilograms can be dominated by weight. [1, pp. 36–37]

$$d(u, v) = \sqrt{\sum_{i=1}^n (u_i - v_i)^2}$$

6.7.1 Convert object features into pairwise distances

Object	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

These are the four points in the slide. The data matrix has four rows and two feature columns. A distance matrix instead has four rows and four columns, both labeled by the objects. Its (i,j) entry compares objects i and j; it is not another measured attribute.

Worked classroom calculation. For p1 and p2, the coordinate differences are –2 and 2, so $d(p1,p2)=\sqrt{(4+4)}=\sqrt{8}\approx 2.828$. For p2 and p3, the differences are –1 and –1, giving $\sqrt{2}\approx 1.414$. For p3 and p4, only x differs, giving distance 2.

Distance	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2.000
p4	5.099	3.162	2.000	0

The diagonal is zero because each point is compared with itself. The matrix is symmetric because reversing a pair does not change Euclidean distance. With four objects, only $4\times 3/2=6$ distinct off-diagonal pairs need to be calculated. The smallest nonzero entry identifies p2 and p3 as the closest pair under this measure.

Supplementary example S8 Scaling changes the comparison. For the earlier people table, P1=(1.79,75) and P2=(1.64,54) have raw distance $\sqrt{(0.15^2+21^2)}\approx 21.0005$. Using the classroom min–max ranges 0.24 m and 24 kg gives differences 0.625 and 0.875 and distance $\sqrt{(0.625^2+0.875^2)}\approx 1.0753$. These values describe different coordinate representations, not a physical separation between people.

The four-point coordinates and distance matrix are classroom material. This section completes the calculations without introducing a clustering algorithm that was deferred to the next class.

6.8 Minkowski distance and different routes

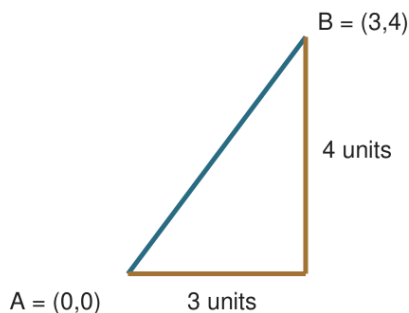
Minkowski distance replaces the square and square root in Euclidean distance with a power r and its reciprocal power. For $r \geq 1$ it defines a family of distances. The lecture discusses $r=1$ and $r=2$ and briefly introduces the limiting case $r \rightarrow \infty$. [1, pp. 38–39; 6]

$$d_r(u, v) = \left(\sum_{i=1}^n |u_i - v_i|^r \right)^{\frac{1}{r}}$$

Choice	Name	Calculation
$r = 1$	Manhattan / city-block / taxicab	Sum absolute coordinate differences
$r = 2$	Euclidean	Square root of sum of squared differences
$r \rightarrow \infty$	Maximum / Chebyshev	Largest absolute coordinate difference

Classroom analogy. A helicopter can follow the direct line between two locations. A taxi on an ideal rectangular street grid travels along the coordinate directions. The first motivates Euclidean distance; the second motivates Manhattan distance. Actual road restrictions can make travel longer than this simplified grid model.

Straight line and axis-aligned route



Euclidean: square root of $(9 + 16) = 5$

Manhattan: $3 + 4 = 7$

Maximum component: $\max(3,4) = 4$

Figure 6.6 Supplementary example S9. The coordinate differences are 3 and 4. The straight route has length 5, an axis-aligned route has length 7, and the maximum coordinate difference is 4.

Apply the rules to the four classroom points

Pair	Absolute differences	Manhattan	Euclidean	Maximum
p1, p2	2, 2	4	$\sqrt{8} \approx 2.828$	2
p2, p3	1, 1	2	$\sqrt{2} \approx 1.414$	1
p3, p4	2, 0	2	2	2

Manhattan distance ties p2–p3 with p3–p4 at 2. Euclidean distance distinguishes them: $\sqrt{2}$ is smaller than 2. Changing the comparison rule can therefore change a nearest-pair decision even though the original points are unchanged.

Why the maximum appears. For differences 3 and 4, write $d_r = 4[1 + (3/4)^r]^{1/r}$. As r grows, the bracketed factor approaches 1, so the distance approaches 4. This worked limit is a supplementary explanation of the briefly introduced maximum-component case.

6.9 Connecting the concepts

Clean matrix → explore distributions → choose objects and features
 → define similarity or distance → form groups → interpret the groups

Visualization helps us inspect the data before defining a comparison. Clustering needs both a meaningful representation and an explicit comparison rule. Cosine emphasizes direction, correlation emphasizes centered linear patterns, and Euclidean or Manhattan distance compares coordinate differences. None answers every biological question equally well.

If the question is...	A relevant starting point
Where are values concentrated or unusual?	Histogram; state the bins
How do location and spread compare?	Box plot; state quartiles and whisker convention
Do two attributes vary together?	Joint plot and an appropriate association measure
Do nonzero profiles have similar proportions?	Cosine similarity
Do profiles have a similar centered linear pattern?	Pearson correlation
How far apart are feature vectors?	A distance with meaningful scaling

Scope boundary: hierarchical clustering was deferred. This edition does not include dendrogram construction, linkage updates, the five-gene hierarchical example, or Mahalanobis distance from later pages of the slide deck.

Sources and example provenance

Page references below use physical PDF pages, because some introductory slides retain Lecture 5 footer labels. Supplementary examples S1–S9 are constructed teaching examples, not empirical findings or quotations from the lecturer. Classroom calculations are distinguished from added explanations.

[1] Li, Y. Clustering. BMEG3105 course slides, lec6-Clustering.pdf. Main scope: PDF pp. 11–39; earlier pages provide recap context.

[2] Li, Y. Data exploration and data cleaning. lec5-Exploration and cleaning.pdf. Corresponding continuation: PDF pp. 50–78.

[3] NIST/SEMATECH e-Handbook. Box Plot. <https://www.itl.nist.gov/div898/handbook/eda/section3/boxplot.htm>. Box structure and alternative whisker conventions.

[4] SciPy documentation. Cosine distance.

<https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.distance.cosine.html>. Distinguishes distance from cosine similarity.

[5] SciPy documentation. Correlation distance.

<https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.distance.correlation.html>. Centered-vector definition.

[6] SciPy documentation. Minkowski distance.

<https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.distance.minkowski.html>. General distance formula.

AI assistance

OpenAI Codex assisted in locating supplementary reference material, developing worked examples, expanding contextual explanations and logical connections, and organizing figures and formatting. Assistance also included drafting and revising explanatory text and checking calculations. The companion working notes follow the same sections with less detail. OpenAI. (2026). Codex. <https://openai.com/codex/>.